

Assignment 4 - nanoGPT Tensor Parallelism on Shakespeare

Daize Dong . Generated 2025-12-05 22:09

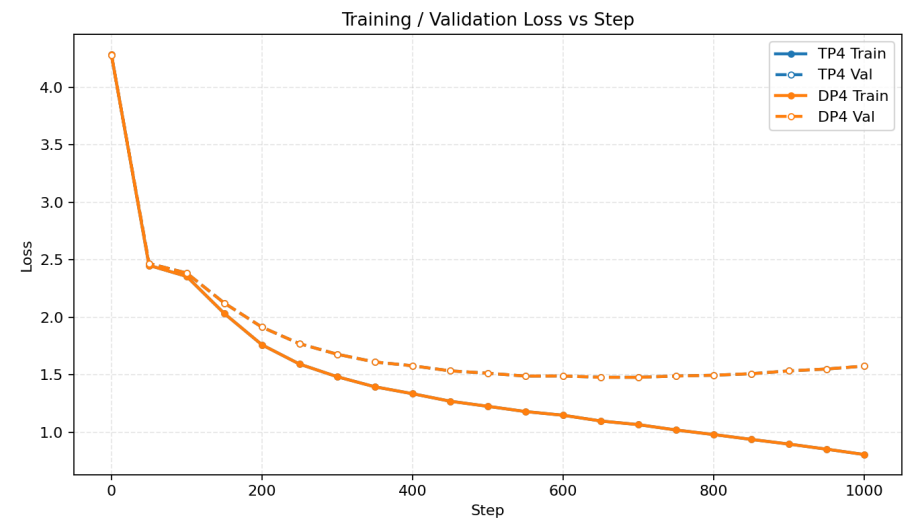
Global Settings

- Global batch size: 512
- Sequence length: 256
- Training Steps: 1000
- Warmup Steps: 20
- Optimizer: AdamW
- Cluster (Delta): 8 A100 GPUs

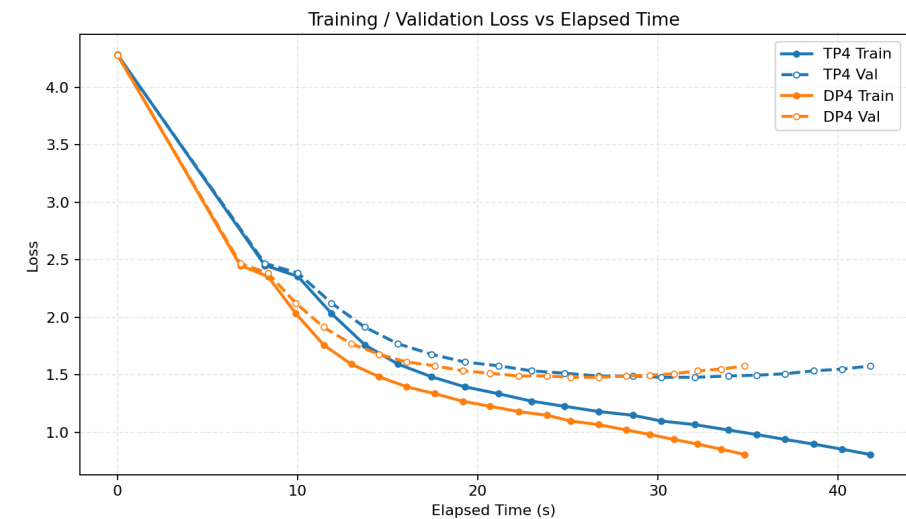
Note that I changed the global batch_size from 64 to 512 to better utilize the GPUs in model parallelism settings. Also, the training steps were reduced from 5000 to 1000 to limit the total runtime for all experiments.

Task 1 & 2: TP4 vs DP4 Comparison

TP4 vs DP4: Loss Curves

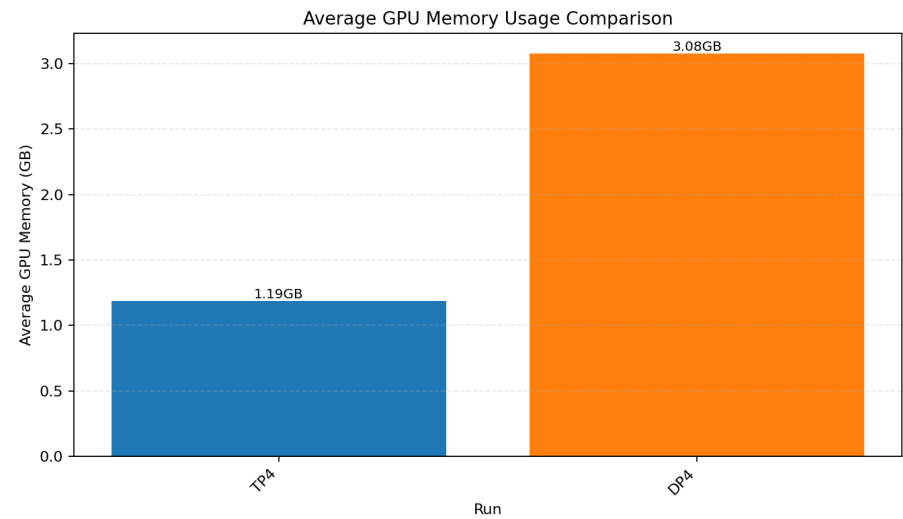


Loss vs steps

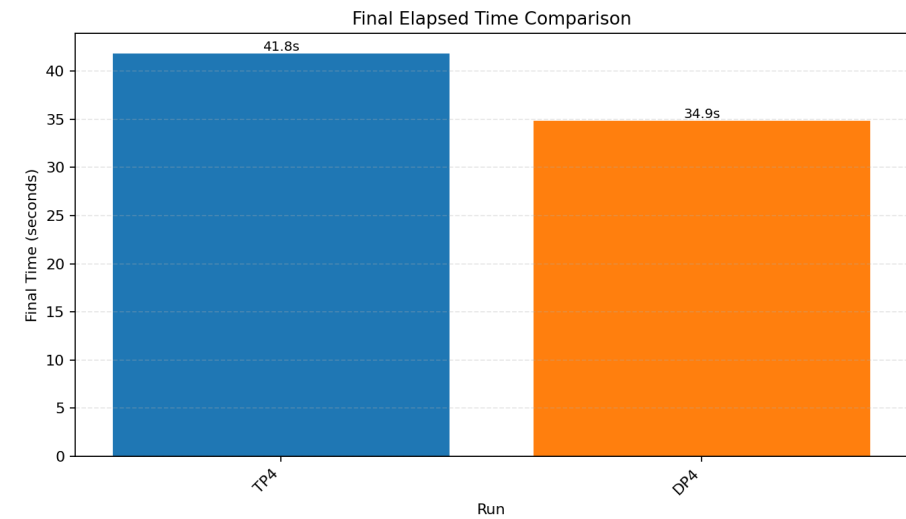


Loss vs time

TP4 vs DP4: Memory and Runtime



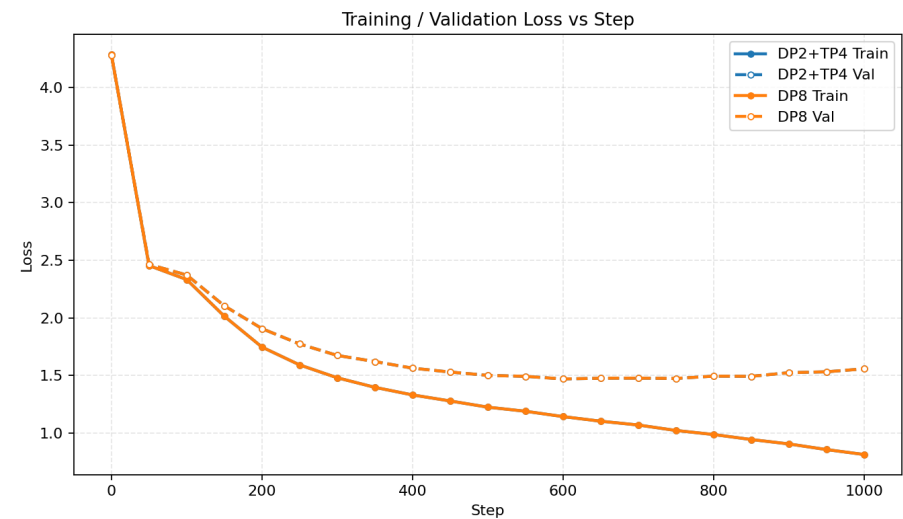
Memory comparison



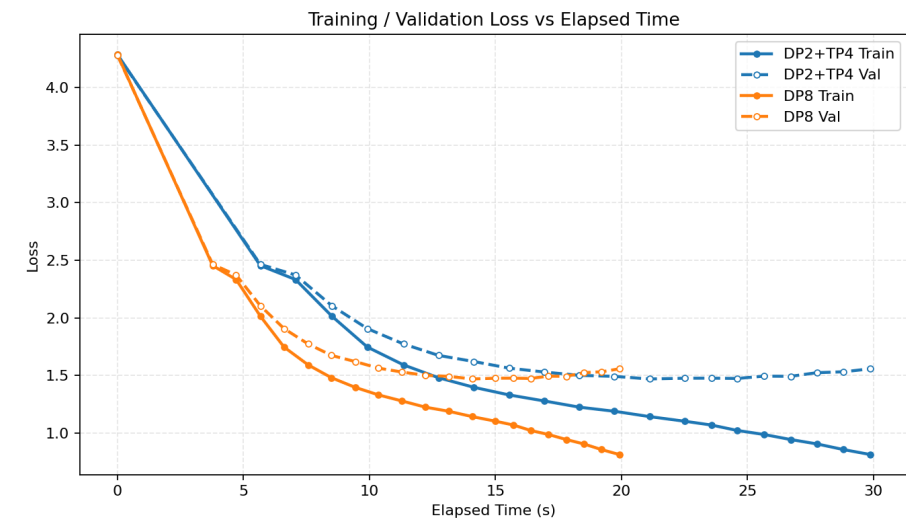
Runtime comparison

Task 3 & 4: DP2+TP4 vs DP8 Comparison

DP2+TP4 vs DP8: Loss Curves

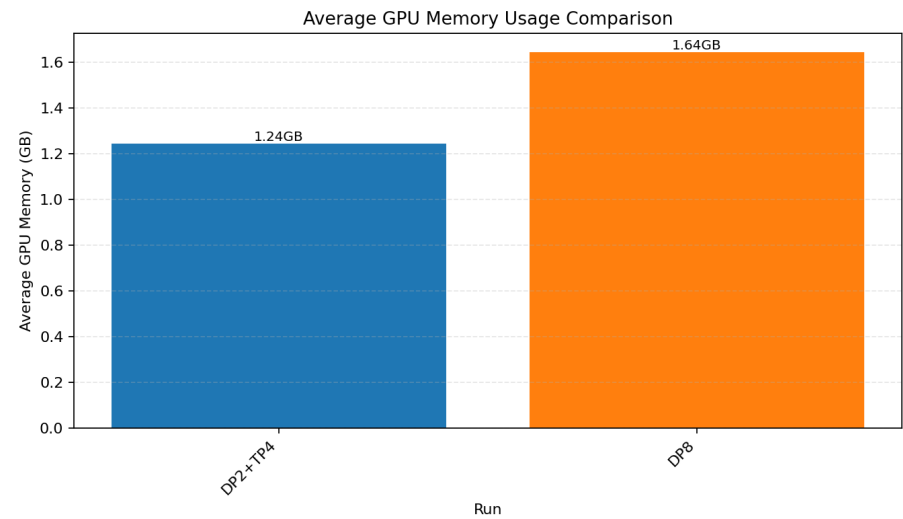


Loss vs steps

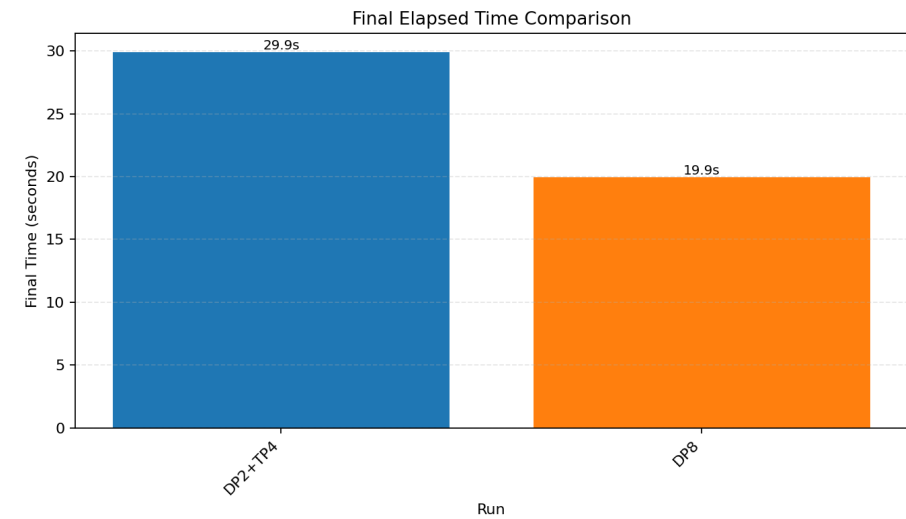


Loss vs time

DP2+TP4 vs DP8: Memory and Runtime



Memory comparison



Runtime comparison